Both R_0 and the herd-immunity threshold need updating



Stacey Smith?

Department of Mathematics and Faculty of Medicine
The University of Ottawa

COVID-19

- COVID-19 resulted in multiple vaccines being released simultaneously
  - e.g., mRNA vaccines, which provide long-lasting immune responses and are potentially much safer than other vaccines
- Having multiple simultaneous vaccines available for a pandemic was unprecedented
- It is likely that more than one vaccine will be available for vaccine-preventable diseases in the future.

Herd immunity

- Herd immunity is a critical determinant for public-health policymakers to contain and potentially eradicate an infectious disease
- It plays an important role in controlling the pandemic once a sufficiently high proportion of the population gains immunity
  - either through vaccination or infection
- The herd-immunity threshold is the proportion of a population that must be vaccinated to stop an infectious disease from spreading.



Vaccinees

- Suppose an infectious disease is distributed homogeneously in the population and a single vaccine provides immunity against it
- Let *p* be the fraction of susceptible individuals who get vaccinated
  - vaccinees are selected randomly with the same probability among susceptible individuals
- We assume the vaccine provides lifelong immunity with no reinfection
  - and protects only a proportion *E* among those infected.

Vaccine effectiveness

- The classic vaccination reproduction number is

$$\mathrm{R}_V = \mathrm{R}_0\left(1 - p\,E\right)$$

  where $E \in [0,1]$ represents vaccine effectiveness
- If the vaccine is perfect and provides 100% immunity, then its effectiveness is $E=1$
  - if it is imperfect and provides only partial immunity, then $E<1$.

Herd-immunity threshold

- The population reaches herd immunity, with incidence of infection declining, if $R_V<1$
- This yields the critical herd-immunity threshold
- Hence a higher proportion needs to be vaccinated if the vaccine is not perfect

$$p^c = \left(1 - \frac{1}{\mathrm{R}_0}\right)\frac{1}{E}$$

  – we set $p^c=1$ if the value is greater than 1.

E=vaccine effectiveness
p=herd-immunity threshold
R_0=reproduction number

Applicability

- It is the most applicable formula used in the history of vaccination
- It represents an upper bound for the threshold under a homogeneity assumption
  - e.g., for age-structured communities with heterogeneous group activities, the disease-induced herd-immunity threshold would be lower than the one under the homogeneity assumption
- It depends critically on R_0.

What about R_0?

- Known as the basic reproduction number
- One of the fundamental concepts in mathematical biology
- Defined as “the average number of secondary infections caused by a single infectious individual during their entire infectious lifetime.”

A brief history of R_0

- Originally developed for demographics (1886)
- Independently studied for malaria (1911,1927)
- Now widely used for infectious disease (1975+)

"One of the foremost and most valuable ideas that mathematical thinking has brought to epidemic theory"

(Heesterbeek & Dietz, 1996).

Definition of R_0

- Expected number of secondary individuals produced by an individual in its lifetime
- However, “secondary” depends on context:
  - mean lifetime reproductive success (demographics and ecology)
  - number of individuals infected within a single infected individual’s entire infectious lifetime (epidemiology)
  - number of newly infected cells produced by a single infected cell (in-host dynamics).

A threshold criterion

- If $R_0<1$, each individual produces, on average, less than one new infected individual...

  ...and hence the disease dies out
- If $R_0>1$, each individual produces more than one new infected individual...

  ...and hence the disease is able to invade the susceptible population
- This allows us to determine the effectiveness of control measures.

R_0 is a threshold

- Predicts whether a disease will become endemic or die out
- If $R_0>1$, then each individual is causing more than one infection, so the disease will take hold
- If $R_0<1$, then the disease will die out
- This threshold is where R_0 is most useful.

R_0 = basic reproductive ratio

Problems with R_0

- R_0 is rarely measured in the field
- R_0 is an individual parameter
- Most models (eg ODEs) deal with populations
- Matching the two $\Rightarrow$ lots of thresholds, not necessarily the true R_0.



Surrogate thresholds

- These surrogate, R_0-like thresholds will also tell us whether a disease will become endemic or die out
- They may be easier to calculate than the true R_0

But…

- We can't compare different diseases.

R_0 = basic reproductive ratio

We can't compare diseases

- Suppose
  - HIV has an R_0 of 3
  - COVID has an R_0 of 5
- Unless they were calculated using the same method, we don't know if COVID is worse than HIV
- All we know is that both will persist
- This is because R_0 is a distance measure and is thus dependent on the metric used.

R_0 = basic reproductive ratio

Anderson & May

$R_0 = \beta\ c\ D$, where

- β = transmission probability
- c = number of contacts per unit time
- D = average time spent infectious

  (= $1/b$ if the recovery rate is b)

This is the method most commonly used by biologists (not always correctly).

Note: This formula only applies if there is no background death rate

R_0 = basic reproductive ratio

A method for determining stability

1. Calculate the disease-free equilibrium
2. Create the Jacobian matrix
3. Evaluate the Jacobian at the equilibrium
4. Find the eigenvalues
5. If the real part of all eigenvalues < 0 $\Longrightarrow$ stable

   If even one eigenvalue has real part > 0 $\Longrightarrow$ unstable
6. Largest eigenvalue $\Longrightarrow$ R_0-like threshold.

R_0 = basic reproductive ratio

The SIS Jacobian

$$\frac{dS}{dt} = bI - aSI$$

$$\frac{dI}{dt} = aSI - bI$$

$$J = \begin{bmatrix} \frac{\partial S'}{\partial S} & \frac{\partial S'}{\partial I} \\ \frac{\partial I'}{\partial S} & \frac{\partial I'}{\partial I} \end{bmatrix}$$

S = Susceptible
I = Infected
a = infection rate
b = recovery rate

The SIS eigenvalues

- The disease-free equilibrium is *(S,I)=(N,0)*

$$J\big|_{(N,0)} = \begin{bmatrix} 0 & b - aN \\ 0 & aN - b \end{bmatrix}$$

- The eigenvalues are

$$\lambda = 0,\ aN - b.$$

S = Susceptible *I = Infected*
N = total pop. *J = Jacobian*
b = recovery rate *a = infection rate*

Stability from largest eigenvalue

The non-constant eigenvalue is $aN-b$

- $aN-b < 0 \Longrightarrow$ equilibrium is stable
- $aN-b > 0 \Longrightarrow$ equilibrium is unstable.

N = total pop.
a = infection rate
b = recovery rate

Defining an R_0-like threshold

Stability if

$$aN - b < 0$$

$$aN < b$$

$$\frac{aN}{b} < 1$$

$\therefore$ Define

$$R_0^{\mathrm{SIS}} = \frac{aN}{b} .$$

N = total pop.
a = infection rate
b = recovery rate

R_0^{SIS} is a threshold, not R_0

- If $R_0^{SIS} < 1$, we have stability of the DFE and hence the disease dies out
- If $R_0^{SIS} > 1$, we have instability of the DFE and hence the disease persists
- But why isn't this R_0?

R_0^{SIS} = SIS model threshold
R_0 = basic reproductive ratio
DFE = Disease-free equilibrium

Answer

1. It isn't unique:
   - $(R_0^{SIS})^2 < 1 \Rightarrow$ disease dies out
   - $(R_0^{SIS})^2 > 1 \Rightarrow$ disease persists
2. There's no reason the largest eigenvalue would inexorably lead to the average number of secondary infections.

R_0^{SIS} = SIS model threshold

What does this mean?

- R_0 is not well defined for ODEs
- But if all we want is a threshold, then this is acceptable
- We have lots of methods of calculating R_0-like thresholds.

R_0 = basic reproductive ratio

SIS endemic equilibrium

- The endemic equilibrium for the SIS model is *(b/a,N-b/a)*
- This only exists if $N-b/a > 0$

$$N > b/a$$

$$aN/b > 1$$

- $\therefore$ Define $R_0^{end} = aN/b$ as before (since if $R_0^{end} > 1$, the disease persists).

N = total pop.
a = infection rate
b = recovery rate

The SIR Jacobian

$$\frac{dS}{dt} = \pi - aSI - \mu S$$

$$\frac{dI}{dt} = aSI - bI - \mu I$$

$$\frac{dR}{dt} = bI - \mu R$$

$$J = \begin{bmatrix} \frac{\partial S'}{\partial S} & \frac{\partial S'}{\partial I} & \frac{\partial S'}{\partial R} \\ \frac{\partial I'}{\partial S} & \frac{\partial I'}{\partial I} & \frac{\partial I'}{\partial R} \\ \frac{\partial R'}{\partial S} & \frac{\partial R'}{\partial I} & \frac{\partial R'}{\partial R} \end{bmatrix}$$

S = Susceptible
I = Infected
R = Recovered
J = Jacobian
π = birth rate
a = infection rate
b = recovery rate
μ = death rate

Jacobian at the DFE

$$J\big|_{(\frac{\pi}{\mu},0,0)} = \begin{bmatrix} -\mu & -\frac{a\pi}{\mu} & 0 \\ 0 & \frac{a\pi}{\mu} - b - \mu & 0 \\ 0 & b & -\mu \end{bmatrix}$$

J = Jacobian
DFE = Disease-free equilibrium
π = birth rate
a = infection rate
b = recovery rate
μ = death rate

Calculating eigenvalues

$$\det(J - \lambda I) = \begin{bmatrix} -\mu - \lambda & -\frac{a\pi}{\mu} & 0 \\ 0 & \frac{a\pi}{\mu} - b - \mu - \lambda & 0 \\ 0 & b & -\mu - \lambda \end{bmatrix}$$

J = Jacobian — *I = identity matrix*
π = birth rate — *a = infection rate*
b = recovery rate — *μ = death rate*

A 3×3 matrix has 3 eigenvalues

- The eigenvalues are thus

$$\lambda = -\mu, -\mu, \frac{a\pi}{\mu} - b - \mu$$

- Rearrange the largest eigenvalue to find

$$R_0^{\mathrm{SIR}} = \frac{a\pi}{\mu(b+\mu)}.$$

π = *birth rate* *a = infection rate*
b = recovery rate μ = *death rate*

SIR endemic equilibrium

- The endemic equilibrium for the SIR model is

$$(\bar{S}, \bar{I}, \bar{R}) = \left(\frac{b+\mu}{a}, \frac{\pi}{b+\mu} - \frac{\mu}{a}, \frac{b\pi}{\mu(b+\mu)} - \frac{b}{a} \right)$$

- The disease persists when $\bar{I} > 0$.

S = Susceptible *I = Infected* *R = Recovered*
π = birth rate *a = infection rate*
b = recovery rate *μ = death rate*

Endemic equilibrium > 0 → $R_0^{end}>1$

- $\bar{I} > 0$:

$$\frac{\pi}{b+\mu} - \frac{\mu}{a} > 0$$

Note: In general, these two methods produce *different* R_0s.

I = Infected
π = birth rate
b = recovery rate
R_0^{end} = endemic threshold
a = infection rate
μ = death rate

Usefulness of the Jacobian

- The Jacobian can determine stability of equilibria
- It can also lead us to R_0-like thresholds
- These determine whether an epidemic will persist or die out.

Other methods

- There are many other methods used to calculate R_0
- Few of them agree with each other
- Even fewer calculate the true R_0.

Survival function

$$R_0 = \int_0^\infty b(a)F(a)da$$

where

- *b(a)* is the average number of newly infected individuals an infectious individual produces per unit time when infected for total time *a*
- *F(a)* is the probability that a newly infected individual remains infectious for at least time *a*.

Infection cycle

Eg malaria:

- $F(a) = \int_0^a \text{prob(human infected at time 0 exists at time t)}$
  $\times \text{prob(human infected for tot. time } t \text{ infects mosquito)}$
  $\times \text{prob(infected mosquito lives to be age } a - t)dt$
- *b(a)* = average number of humans newly infected by a mosquito which has been infected for time *a*
- This yields the total number of humans produced by one infected human.

The “gold standard” of R_0

- Takes into account the complete cycle
- Not restricted to ODEs
- But can get cumbersome for multiple states (eg schistosomiasis)

Applications: Scrapie in Cyprus (Gravenor *et al.*, 2004),
Dengue in Brazil (Luz *et al.*, 2003).





The next-generation method

- A general method for deriving R_0 when the population is divided into discrete, disjoint cases
- Can be used for models with underlying age structure or spatial structure.



Calculating the next-generation R_0

Let

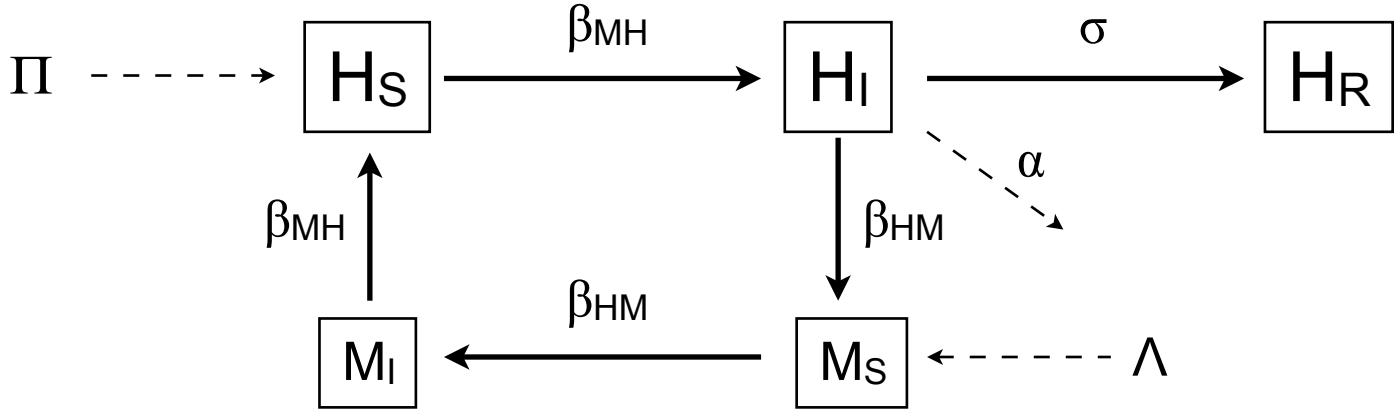
- F_i be the rate of appearance of new infections in compartment *i*
- V_i be the transfer of individuals out of compartment *i* by all other means
- x_0 be the disease-free equilibrium
- R_0 is the real part of the largest eigenvalue of

Matrix of partial derivatives

$$\left[\frac{\partial F_i(x_0)}{\partial x_j}\right] \cdot \left[\frac{\partial V_i(x_0)}{\partial x_j}\right]^{-1}$$

Matrix of partial derivatives.

A malaria model



$$
\begin{aligned}
H_S' &= \Pi - \beta_{MH} M_I H_S - \mu_H H_S \\
H_I' &= \beta_{MH} M_I H_S - (\mu_H + \alpha + \sigma) H_I \\
H_R' &= \sigma H_I - \mu_H H_R \\
M_S' &= \Lambda - \beta_{HM} M_S H_I - \mu_M M_S \\
M_I' &= \beta_{HM} M_S H_I - \mu_M M_I
\end{aligned}
$$

Infected humans

Infected mosquitos.

Eg malaria

$$
\begin{aligned}
H_I' &= \beta_{MH} M_I H_S - (\mu_H + \alpha + \sigma) H_I \\
M_I' &= \beta_{HM} M_s H_I - \mu_M M_I
\end{aligned}
$$

Eg malaria

$$
\begin{aligned}
H_I' &= \beta_{MH} M_I H_S - (\mu_H + \alpha + \sigma) H_I \\
M_I' &= \beta_{HM} M_s H_I - \mu_M M_I \\
F &= \begin{pmatrix} 0 & \beta_{MH} H_S(0) \\ \beta_{HM} M_S(0) & 0 \end{pmatrix}
\end{aligned}
$$

Eg malaria

$$
\begin{aligned}
H_I' &= \beta_{MH} M_I H_S - (\mu_H + \alpha + \sigma) H_I \\
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V &= \begin{pmatrix} \mu_H + \alpha + \sigma & 0 \\ 0 & \mu_M \end{pmatrix}
\end{aligned}
$$

Eg malaria

$$
\begin{aligned}
H_I' &= \beta_{MH} M_I H_S - (\mu_H + \alpha + \sigma) H_I \\
M_I' &= \beta_{HM} M_s H_I - \mu_M M_I \\
F &= \begin{pmatrix} 0 & \beta_{MH} H_S(0) \\ \beta_{HM} M_S(0) & 0 \end{pmatrix} \\
V &= \begin{pmatrix} \mu_H + \alpha + \sigma & 0 \\ 0 & \mu_M \end{pmatrix} \\
R_{0,N} &= \sqrt{\frac{\beta_{MH}\beta_{HM} H_S(0) M_S(0)}{(\mu_H + \alpha + \sigma)\mu_M}}
\end{aligned}
$$

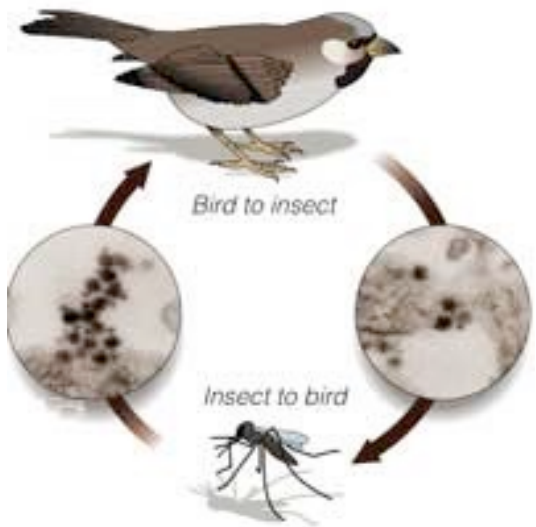
Eg malaria

$$
\begin{aligned}
H_I' &= \beta_{MH} M_I H_S - (\mu_H + \alpha + \sigma) H_I \\
M_I' &= \beta_{HM} M_s H_I - \mu_M M_I \\
F &= \begin{pmatrix} 0 & \beta_{MH} H_S(0) \\ \beta_{HM} M_S(0) & 0 \end{pmatrix} \\
V &= \begin{pmatrix} \mu_H + \alpha + \sigma & 0 \\ 0 & \mu_M \end{pmatrix} \\
R_{0,N} &= \sqrt{\frac{\beta_{MH}\beta_{HM} H_S(0) M_S(0)}{(\mu_H + \alpha + \sigma)\mu_M}} \\
&= \sqrt{R_{0,S}} \text{ (the } R_0 \text{ from the survival method).}
\end{aligned}
$$

Difference: definition of a “generation”

- Survival method: the number of infected humans produced by a single infected human
- Next-generation: the number of infected *mosquitos* produced by a single human

Applications: West Nile virus (Wonham *et al.*, 2004),
Mad cow (de Koeijer *et al.*, 2003).





Threshold criterion

- The true R_0 is clearly a threshold
- Other expressions are also thresholds
- eg $(R_0)^k$ for $k > 0$
- However, these thresholds are not necessarily the average number of secondary infections

The Jacobian method again

- Stability of the disease-free equilibrium is determined by the largest eigenvalue of the Jacobian
- This is a threshold condition that can derive an $R_{0,J}$
- May not produce a biologically meaningful value

Applications: malaria immunology (Tumwiine et al, 2007).



Eg malaria again, using the Jacobian

$$J(x_0) = \begin{bmatrix} -(\mu_H + \alpha + \sigma) & \beta_{MH} H_S(0) \\ \beta_{HM} M_S(0) & -\mu_M \end{bmatrix}$$

Eg malaria again, using the Jacobian

$$J(x_0) = \begin{bmatrix} -(\mu_H + \alpha + \sigma) & \beta_{MH} H_S(0) \\ \beta_{HM} M_S(0) & -\mu_M \end{bmatrix}$$

$$\lambda_{\max} = \frac{1}{2}\Bigg\{ -\mu_M - \mu_H - \alpha - \sigma + \sqrt{\left[\mu_M - (\mu_H + \alpha + \sigma)\right]^2 + \frac{4\beta_{HM}\beta_{MH} H_S(0) M_S(0)}{\mu_H \mu_M}} \Bigg\}$$

Eg malaria again, using the Jacobian

$$J(x_0) = \begin{bmatrix} -(\mu_H + \alpha + \sigma) & \beta_{MH} H_S(0) \\ \beta_{HM} M_S(0) & -\mu_M \end{bmatrix}$$

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$$R_{0,J} = \sqrt{\frac{\left[\mu_M - (\mu_H + \alpha + \sigma)\right]^2 + \frac{4\beta_{HM}\beta_{MH} H_S(0) M_S(0)}{\mu_H \mu_M}}{(\mu_M + \mu_H + \alpha + \sigma)^2}}$$

Eg malaria again, using the Jacobian

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$$R_{0,J} = \sqrt{\frac{\left[\mu_M - (\mu_H + \alpha + \sigma)\right]^2 + \frac{4\beta_{HM}\beta_{MH} H_S(0) M_S(0)}{\mu_H \mu_M}}{(\mu_M + \mu_H + \alpha + \sigma)^2}}$$

$$\neq R_{0,S},\ R_{0,N}.$$

Constant term of the characteristic polynomial

- Malaria again:

Constant term of the characteristic polynomial

- Malaria again:

$$J(x_0) = \begin{bmatrix} -(\mu_H + \alpha + \sigma) & \beta_{MH} H_S(0) \\ \beta_{HM} M_S(0) & -\mu_M \end{bmatrix}$$

Constant term of the characteristic polynomial

- Malaria again:

$$J(x_0) = \begin{bmatrix} -(\mu_H + \alpha + \sigma) & \beta_{MH} H_S(0) \\ \beta_{HM} M_S(0) & -\mu_M \end{bmatrix}$$

$$\begin{aligned} \det(J - \lambda I) &= \lambda^2 + (\mu_M + \mu_H + \alpha + \sigma)\lambda \\ &\quad + \mu_M(\mu_H + \alpha + \sigma) - \beta_{MH}\beta_{HM} M_S(0) H_S(0) \end{aligned}$$

Constant term of the characteristic polynomial

- Malaria again:

$$J(x_0) = \begin{bmatrix} -(\mu_H + \alpha + \sigma) & \beta_{MH} H_S(0) \\ \beta_{HM} M_S(0) & -\mu_M \end{bmatrix}$$

$$\begin{aligned} \det(J - \lambda I) &= \lambda^2 + (\mu_M + \mu_H + \alpha + \sigma)\lambda \\ &\quad + \mu_M(\mu_H + \alpha + \sigma) - \beta_{MH}\beta_{HM} M_S(0) H_S(0) \end{aligned}$$

$$R_{0,C} = \frac{\beta_{MH}\beta_{HM} M_S(0) H_S(0)}{\mu_M(\mu_H + \alpha + \sigma)}$$

Constant term of the characteristic polynomial

- Malaria again:

$$J(x_0) = \begin{bmatrix} -(\mu_H + \alpha + \sigma) & \beta_{MH} H_S(0) \\ \beta_{HM} M_S(0) & -\mu_M \end{bmatrix}$$

$$\begin{aligned} \det(J - \lambda I) &= \lambda^2 + (\mu_M + \mu_H + \alpha + \sigma)\lambda \\ &\quad + \mu_M(\mu_H + \alpha + \sigma) - \beta_{MH}\beta_{HM} M_S(0) H_S(0) \end{aligned}$$

$$R_{0,C} = \frac{\beta_{MH}\beta_{HM} M_S(0) H_S(0)}{\mu_M(\mu_H + \alpha + \sigma)}$$

$$= R_{0,S}$$

Constant term of the characteristic polynomial

- Malaria again:

$$J(x_0) = \begin{bmatrix} -(\mu_H + \alpha + \sigma) & \beta_{MH} H_S(0) \\ \beta_{HM} M_S(0) & -\mu_M \end{bmatrix}$$

$$\begin{aligned} \det(J - \lambda I) &= \lambda^2 + (\mu_M + \mu_H + \alpha + \sigma)\lambda \\ &\quad + \mu_M(\mu_H + \alpha + \sigma) - \beta_{MH}\beta_{HM} M_S(0) H_S(0) \end{aligned}$$

$$R_{0,C} = \frac{\beta_{MH}\beta_{HM} M_S(0) H_S(0)}{\mu_M(\mu_H + \alpha + \sigma)}$$

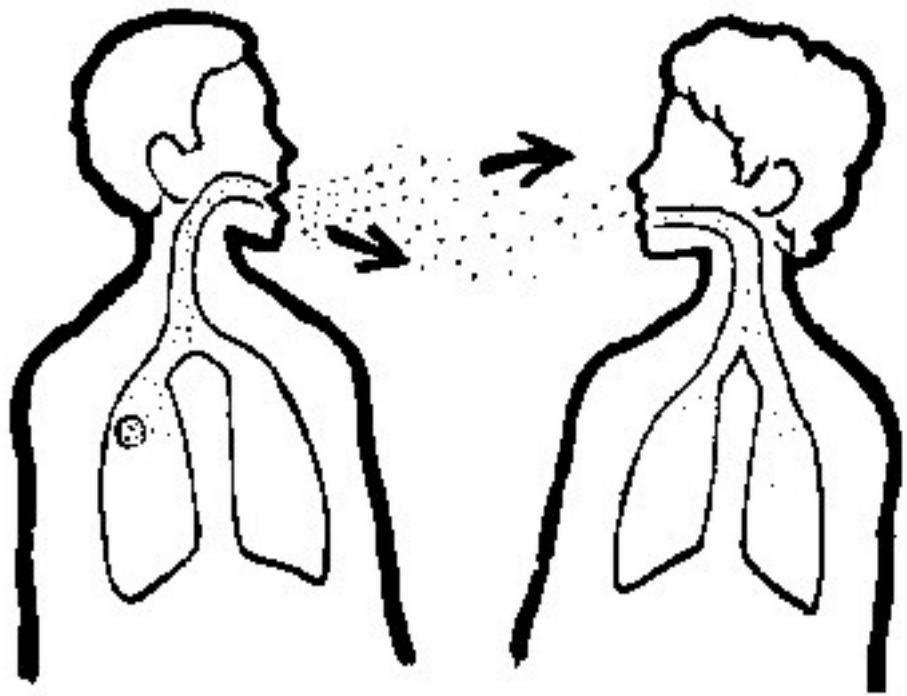
$$= R_{0,S}$$

$$\neq R_{0,J}, R_{0,N}.$$

Constant term of the polynomial

- Easier to calculate than most other methods
- Especially useful for calculating different R_0s corresponding to different strains of a disease

Applications: tuberculosis in hot zones (Blower & Chou, 2004).



R_0 Summary

- Many threshold parameters
- Few of these reliably calculate the average number of secondary infections
- R_0 can't be used to compare different diseases
- However, multiple methods provide a variety of ways to determine thresholds.

What about the herd-immunity?

- R_0 is a critical component of calculating the herd-immunity threshold
- However, there are further issues here
- This herd-immunity formula is based on an assumption of homogeneity in the infected population
  - and also the assumption that there is only one available vaccine
  - and one strain of the disease
- Neither of these things were true during COVID-19.

Multiple vaccines

- Suppose there are k>1 effective vaccines for a single disease
- Let E_j be the effectiveness of the *j*-th vaccine
- The most favourable strategy is to use a single vaccine with the largest effectiveness
  - however, this vaccine may not be fully available in time
- A combination of vaccines may thus be the optimal strategy
  - the least favourable strategy is a single vaccine with the smallest effectiveness.

Multi-vaccine reproduction number

- Let p_j be the proportion of susceptibles who receive the *j*-th vaccine
- Let S be an allocation strategy with k proportions $(p_1,p_2,\ldots,p_k)$ such that $\sum_{j=1} p_j \leq 1$
- Suppose the populations receiving different vaccines are disjoint
  - a simplifying mathematical assumption
- The vaccine reproduction number with this multi-vaccine strategy is

$$\mathrm{R}_V^{\mathrm{S}} = \mathrm{R}_0\left(1 - \sum_{j=1}^{k} p_j E_j\right).$$

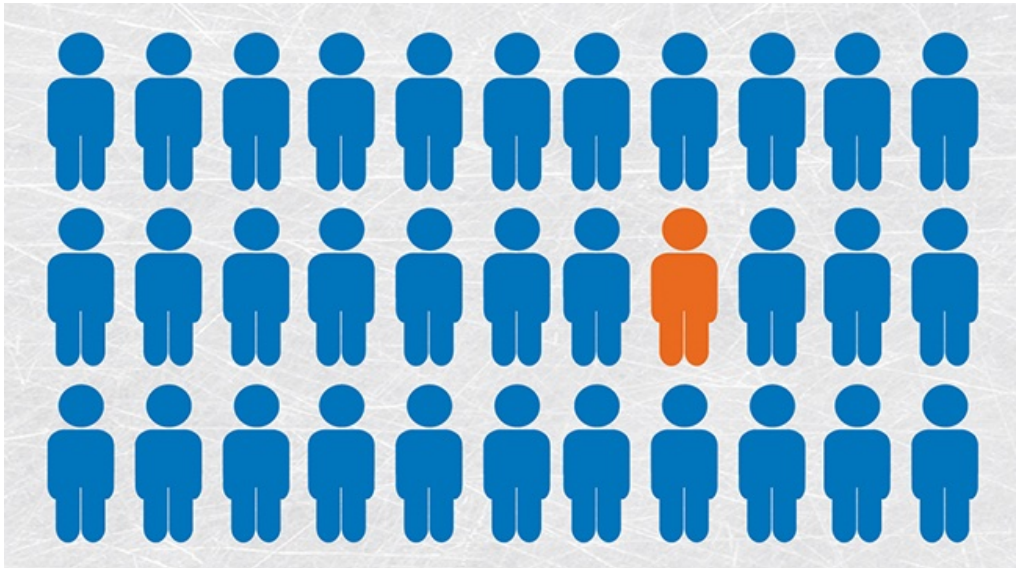
E=vaccine effectiveness
p=herd-immunity threshold
R_0=reproduction number

Goal

- Our goal is to find a critical lower bound for the total proportion of the population

$$p_{\mathrm{S}} = \sum_{j=1}^{k} p_j$$

that must be vaccinated under strategy S to achieve $\mathrm{R}_V^S < 1$.



p=herd-immunity threshold
R_0=reproduction number

Range

- The lower bound of p_S is between the critical herd-immunity thresholds for the least and most favourable strategies

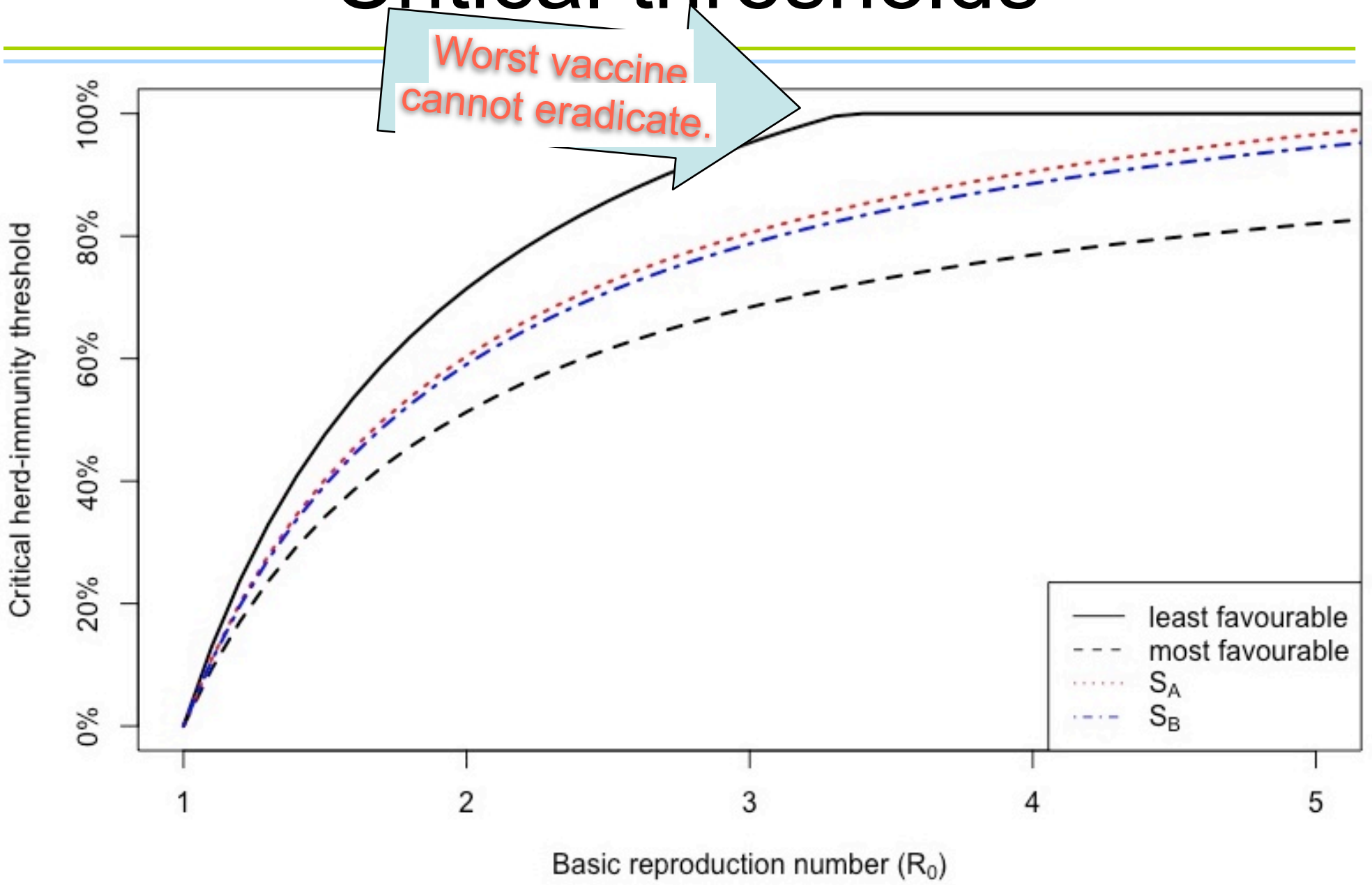
$$\min_j p_j^c \leq p_S \leq \max_j p_j^c$$

where p_j^c is the critical herd-immunity threshold if only the *j*-th vaccine were used.



p=herd-immunity threshold

Critical thresholds



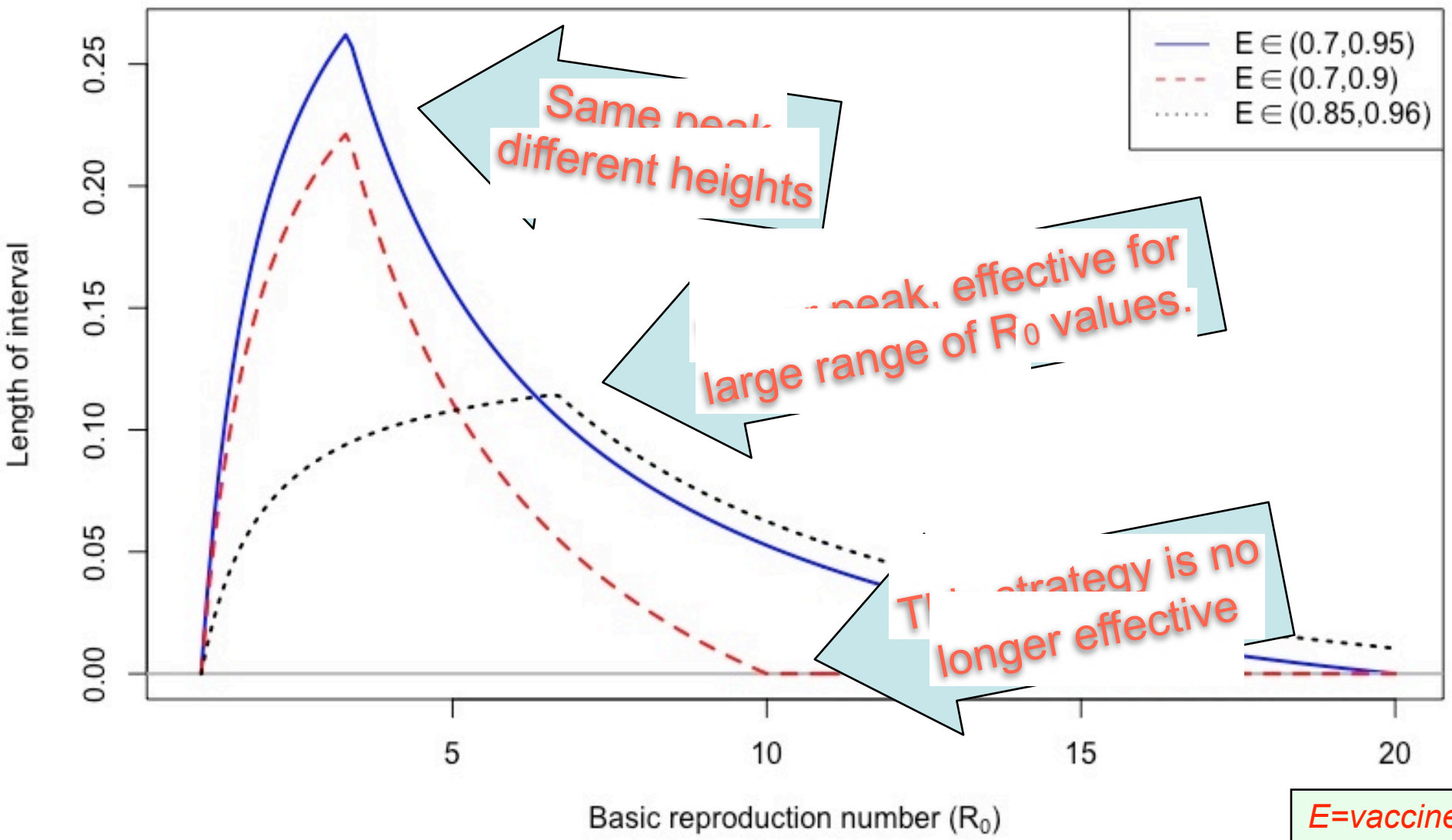
Least – most interval

- Let δ be the length of the interval

  representing the difference between the least favourable and the most favourable vaccination strategies
- Hence $\delta = \max_j p_j^c - \min_j p_j^c$
- δ is increasing for small values of R_0
- It is maximised at a point where the least favourable strategy is not effective any more.

p=herd-immunity threshold
R_0=reproduction number

Length of interval



E=vaccine effectiveness

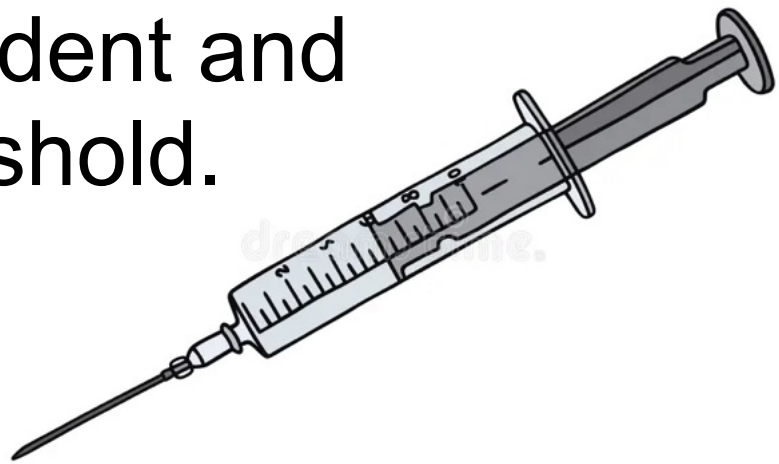
Max vs min δ

- The peak of the curve is the last point where all vaccines are effective
- i.e., at $\mathrm{R}_0 = 1/(1 - \min_{j=1}^{k} E_j)$
- δ is minimised to zero at $\mathrm{R}_0 = 1/(1 - \max_{j=1}^{k} E_j)$
- This is the last point at which the most favourable strategy is effective
- After this point, δ is zero
  - achieving vaccine-induced herd immunity is then impossible under any combination of vaccines.

E=vaccine effectiveness
R_0=reproduction number
δ=interval length

Time-dependent threshold

- δ can be interpreted as the effect of changing the share of a vaccine on the herd-immunity threshold
- Acceptance rates of vaccines are different
  - and their shares may change during a vaccination program
- This results in a time-dependent and variable herd-immunity threshold.



δ=interval length

Multi-vaccine strategies

- We can propose effective multi-vaccine strategies if the reproduction number is close to the peak of δ
- Conversely, improving a vaccination strategy is difficult for values of R_0 when δ is small
- Hence, replacing a vaccine with one that has a better effectiveness may not have a substantial effect on the herd-immunity threshold.



R_0=reproduction number
δ=interval length

α-coefficients

- Suppose $\alpha_j = p_j / p_j^c$, where p_j^c is the critical threshold for the *j*-th population if it were used for the whole population
  - a measure of how much that vaccine is contributing relative to its strength
- The coefficients $\alpha = (\alpha_1, \ldots \alpha_k)$ describe the vaccine strategy
  - note that these coefficients are not the fraction of the population that gets that vaccine.

p=herd-immunity threshold

Reaching herd immunity

- The total proportion of the population that must be vaccinated under strategy S is

$$p_{\mathrm{S}}^{c} = \left(1 - \frac{1}{\mathrm{R}_0}\right) \frac{\sum_{j=1}^{k} \alpha_j / E_j}{\sum_{j=1}^{k} \alpha_j}$$

- Vaccine-induced herd immunity can be achieved if $\sum_{j=1}^{k} \alpha_j > 1$
- This is equivalent to $R_V<1$.

E=vaccine effectiveness
p=herd-immunity threshold
R_0=reproduction number
α=strategy coefficients



Changing vaccination plans

- Many social, political and economic factors can affect a vaccination program
- A vaccination process could be stopped at any time
  - e.g., due to new evidence about side effects
- Suppose a vaccination process started with k vaccines under strategy S_1 with proportions $(p_{11}^*,\ldots,p_{1k}^*)$
- After a while, it stops, and a new strategy occurs (with or without a pause).

p=herd-immunity threshold

Second strategy

- Suppose the second step contains k′ vaccines with effectivenesses $E_{21},\ldots,E_{2k'}$ under a new strategy S_2 with proportions $(p_{21},\ldots,p_{2k'})$
- The reproduction number is

$$\mathrm{R}_V^{\mathrm{S}_2} = \mathrm{R}_0 \left(1 - q - \sum_{j=1}^{k'} p_{2j} E_{2j}\right)$$

where $q = \sum_{j=1}^{k} p_{1j}^{*} E_{1j}$ is an approximation of the proportion of individuals immunised in the first strategy

  – by natural infection or vaccination.

E=vaccine effectiveness
p=herd-immunity threshold
R_0=reproduction number

Two-step herd immunity

- Note that *q* can be considered as the proportion of individuals who are immunised naturally by infection
  - and if infected people are excluded from the vaccination program
- Setting $p_{2j}=\alpha_{2j}p_{2j}^{c}$ implies that herd immunity can be reached if

$$\sum_{j=1}^{k'} \alpha_{j2} > 1 - \frac{q}{1-\frac{1}{R_0}}.$$

p=herd-immunity threshold
R_0=reproduction number
α=strategy coefficients

Multi-step processes

- The leads to a smaller level of vaccination coverage in order to reach herd immunity
- Note that this result can be extended to cases where the vaccination process has more than two steps
- Next, we examine optimal vaccination strategies when *k* vaccines are available
  - with coefficients $\alpha_1,\ldots,\alpha_k$
  - and effectivenesses $E_1,\ldots,E_k$.

E=vaccine effectiveness
α=strategy coefficients

Vaccine utility

- Let *u* be a positive variable that represents the utility of the vaccines
- If $u_i < u_j$, then the *j*-th vaccine has a larger coefficient than the *i*-th vaccine
- We can thus define $\alpha_j = \frac{u_j^n}{\sum_{j=1}^{k} u_j^n}$ where n≥0 is a shrinkage parameter
- By controlling *n*, some coefficients can be shrunk to zero
  - the corresponding vaccines are then removed from the program.

α=strategy coefficients

Some ranking methods

A. When there is no prior information to sort available vaccines, we set all coefficients to $\alpha_j=1/k$
   - equivalent to a classic single-vaccine strategy

B. If vaccines are sorted by effectiveness, set $u_j=E_j$
   - used when all vaccines are effective, but the variation between effectivenesses is large.

E=vaccine effectiveness
α=strategy coefficients
k=number of vaccines

More ranking methods

C. If reducing side effects is critical, we must maximise the proportion of individuals who remain both unvaccinated and uninfected
   - this proportion is known as the herd effect
   - set $u_j=1-p_j^c$
   - if a vaccine is not effective, its coefficient is zero

D. Vaccines may be ranked on multiple criteria, such as length of immunity, vaccine availability, cost etc
   - if r_j is the rank of the *j*-th vaccine, such that larger ranks are more important, set $u_j=r_j$.

p=herd-immunity threshold
u=vaccine utility

Optimisation

- Some people cannot be vaccinated
  - e.g., pregnant women, immunocompromised individuals, young children
- The ideal vaccination policy is the one with the lowest herd-immunity threshold, which allows the most people to avoid infection without vaccination
- This optimisation problem has been previously examined for a single vaccine
  - but not for multiple vaccines.

Objective functional

- Let *S* be a strategy with coefficients satisfying $\sum_{j=1}^{k} \alpha_j > 1$
- The optimal strategy achieves herd immunity while maximising the number of people who escape infection without vaccination
- This is equivalent to minimising

$$\sum_{j=1}^{k} \alpha_j / E_j$$

subject to the above constraint.

E=vaccine effectiveness
α=strategy coefficients

Cost effectiveness

- Optimising total cost of vaccination is an important consideration for policymakers
  - health budgets are often limited
- Let *N* be the population size and c_j be the cost of the *j*-th vaccine per individual
- A lower bound for the total cost under strategy *S* is

$$N(1-\frac{1}{\mathrm{R}_0})\sum_{j=1}^{k}\alpha_j c_j/E_j.$$

E=vaccine effectiveness
R_0=reproduction number
α=strategy coefficients

Convex optimisation

- Assuming *N* and R_0 are constant, we can estimate the vaccine weights by minimising the cost and maximising the herd effect simultaneously
- This yields the convex optimisation problem

$$\min_{\alpha} \sum_{j=1}^{k} \alpha_j \left(\frac{c_j + 1}{E_j} \right) \quad \text{such that} \quad \sum_{j=1}^{k} \alpha_j > 1$$

where c_j and E_j are known constant values.

E=vaccine effectiveness
R_0=reproduction number
α=strategy coefficients
c=costs

Worked example

- Suppose we have a disease with $R_0=2.5$
- Six vaccines are available, with efficacies $E_1=0.8$, $E_2=0.85$, $E_3=0.9$, $E_4=0.94$, $E_5=0.95$ and $E_6=0.95$
- Let S_{Vj} be the single-vaccine strategy with only the *j*-th vaccine used
- Suppose S_A, S_B, S_C and S_D are multi-vaccine strategies based on a ranking method
- We add a shrinkage factor $n \in \{1,10,20\}$.

Multiple vaccina egies

	Strategy	Vaccines(Efficacy) $V_1(0.80)$	$V_2(0.85)$	$V_3(0.90)$	$V_4(0.94)$	$V_5(0.95)$	$V_6(0.95)$	p_S^f	$\varepsilon(S)$	$\bar{E}_S$
Single-vaccine	S_{V_1}	1	0	0	0	0	0	0.750	0.000	0.80
	S_{V_2}	0	1	0	0	0	0	0.706	0.373	0.85
	S_{V_3}	0	0	1	0	0	0	0.667	0.704	0.90
	S_{V_4}	0	0	0	1	0	0	0.638	0.943	0.94
		0	0	0			0	0.632	1.000	0.95
			0	0				0.632	1.000	0.95
			1/6	1/6				0.671	0.672	0.895
Multi-vaccine strategies			0.15	0.17	0.18	0.19	0.19	0.667	0.691	0.898
	S_B^{10}	0.04	0.08	0.15	0.23	0.25	0.25	0.650	0.850	0.924
	S_B^{20}	0.01	0.03	0.10	0.25	0.30	0.30	0.640	0.926	0.937
	S_C^1	0.12	0.15	0.17	0.18	0.19	0.19	0.665	0.722	0.903
	S_C^{10}	0.01	0.03	0.11	0.25	0.30	0.30	0.640	0.926	0.937
	S_C^{20}	0.00	0.00	0.05	0.25	0.35	0.35	0.635	0.966	0.944
	S_D^1	0.05	0.10	0.14	0.19	0.26	0.26	0.650	0.838	0.922
	S_D^{10}	0.00	0.00	0.00	0.02	0.49	0.49	0.632	1.000	0.950
	S_D^{20}	0.00	0.00	0.00	0.00	0.50	0.50	0.632	1.000	0.950

Single
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Strategy B uses
vaccines, but
reduces reliance on the
weak ones

Uniform distribut
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best
some of the
second best.

Vaccine allocation

- Making decisions for vaccine allocation is a complex problem
- It depends on multiple priorities
- Our table illustrates plausible strategies that satisfy logistic, financial and social conditions of the public-health system.



Multiple variants

- When several variants of a virus exist, adjustments are needed to estimate the herd-immunity threshold
- First, suppose only a single vaccine exists to control a virus with *m* variants
- The proportions of each variant are $\pi_1,\ldots,\pi_m$, with

$$\sum_{j=1}^{m} \pi_j = 1.$$



One vaccine, multiple variants

- Let $E_1,\ldots,E_m$ be the effectivenesses of the vaccine against the variants
- The vaccination reproduction number is

$$\bar{\mathrm{R}}_V = \mathrm{R}_0\,(1 - p\bar{E})$$

where R_0 is the reproduction number for all variants and

$$\bar{E} = \sum_{j=1}^{m} \pi_j E_j$$

is the average effectiveness of the vaccine against the virus in the population.

E=vaccine effectiveness
p=herd-immunity threshold
R_0=reproduction number
π=proportions of variants

Multiple vaccines, multiple variants

- Now suppose *k* vaccines are available for m variants
- The effectiveness of the *i*-th vaccine against the *j*-th variant is E_{ij}
- The vaccination reproduction number is now

$$\bar{\mathrm{R}}_V = \mathrm{R}_0 \left(1 - \sum_{j=1}^{k} p_i \bar{E}_i\right)$$

where the average effectiveness of the *i*-th vaccine is

$$\bar{E}_i = \sum_{j=1}^{m} \pi_j E_{ij}.$$

E=vaccine effectiveness
p=herd-immunity threshold
R_0=reproduction number
π=proportions of variants

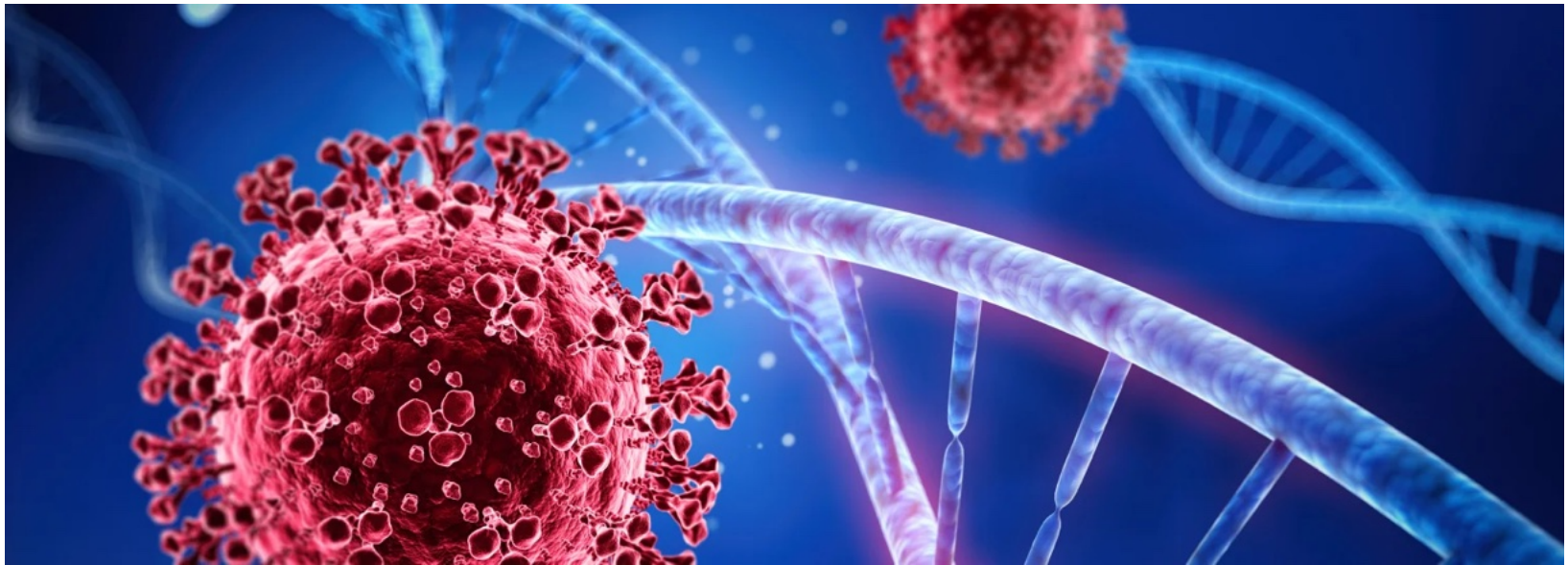
Application to COVID-19

- COVID-19 has several variants
- However, we also have multiple vaccines, with differing effectivenesses
- We applied several strategies to the situation in 2021 (before Omicron)
- Strains include Alpha, Beta, Gamma and Delta
- Vaccines include Pfizer, Moderna, Vaxzevria (formerly AstraZeneca) and unreported
  - we considered one vs two doses.

COVID-19 in Canada in 2021

	Vaccine type	Percentage (p_i)	Effectiveness against variants (E_{ij})				$\bar{E}_i$
			Original strain	Alpha	Beta/Gamma	Delta	
Partial (single-dose) vaccination	Pfizer	5.40%	80%	66%	60%	56%	71%
	Moderna	2.20%	80%	83%	77%	72%	80%
	Vaxzevria	0.12%	70%	64%	48%	67%	65%
	Vaccine not reported	3.			65%	61%	74%
Full (two-doses) vaccination	Pfizer	28		9%	84%	87%	91%
	Moderna	8.		92%	77%	72%	89%
	Vaxzevria	0		74%	48%	67%	75%
	Vaccine not reported	1		89%	82%	83%	90%
	mixed	9.7	85%	74%	48%	67%	75%
	Variant prevalence (π_j)	–	47%	31%	15%	6%	–

Th ss is not always predictable.



Modified reproduction number

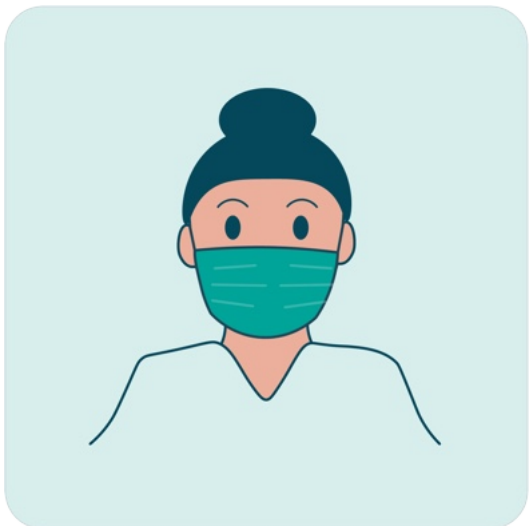
- If only the original variant exists in the population, with R_0=2.5, then the vaccination reproduction number is

$$\mathrm{R}_V = \mathrm{R}_0 \left(1 - \sum_{\text{Partial}} p_i E_{i1} - \sum_{\text{Full}} p_i E_{i1}\right) = 0.8817$$

- However, if several variants coexist according to Table 2, then our modified reproduction number is

$$\bar{\mathrm{R}}_V = \mathrm{R}_0 \left(1 - \sum_{\text{Partial}} p_i \bar{E}_i - \sum_{\text{Full}} p_i \bar{E}_i\right) = 0.9892$$

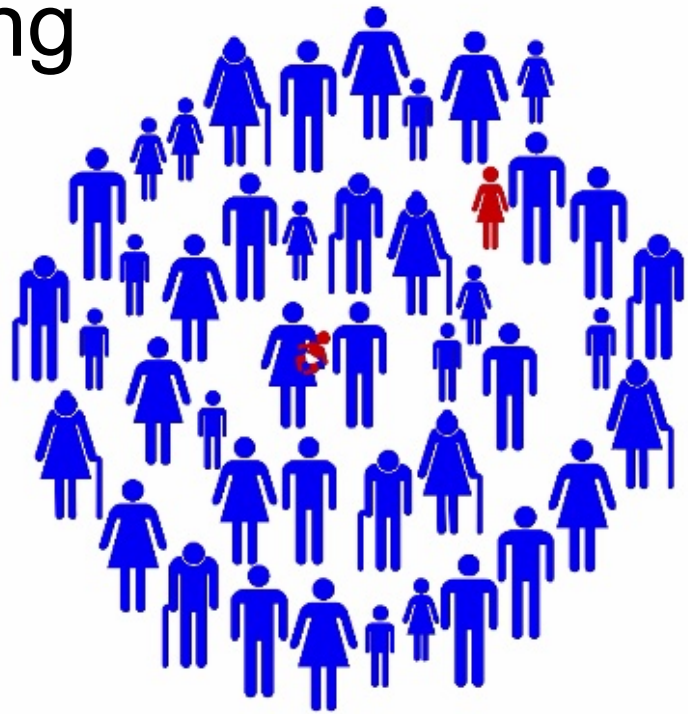
- This is a 12% increase.



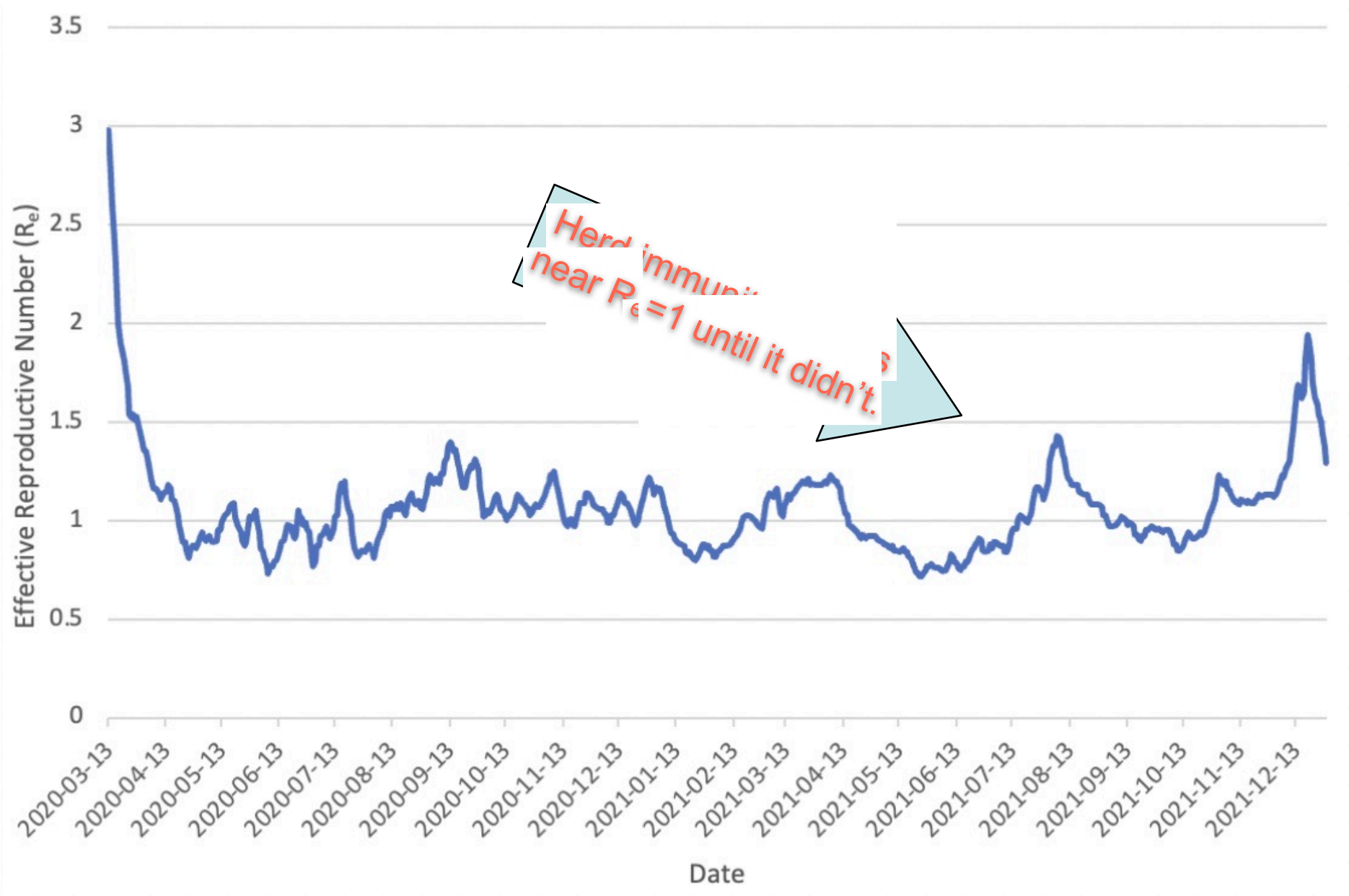
E=vaccine effectiveness R_0=reproduction number
p=herd-immunity threshold

Underestimation

- Hence if we do not modify our calculations in the presence of emerging variants, we would underestimate the reproduction number
- This can lead to underestimating the herd immunity threshold
  - we saw the rapid outbreak of the Omicron strain in an otherwise mostly vaccinated population.



Reproduction numbers in Ontario



Conclusion

- We recently faced the unprecedented situation of a pandemic with multiple simultaneous vaccines and multiple co-circulating variants
- Updating the herd-immunity threshold for multiple vaccines and multiple variants will hasten eradication and save lives
- Can we use graph-theoretical methods to calculate the updated herd-immunity thresholds in more intuitive ways?

Project

- de-Camino-Beck, et al. used a graph-theoretic method to calculate R_0
  - it inherited its structure from the next-generation method
  - and also the same issues

de-Camino-Beck T, Lewis MA, van den Driessche P. (2009) A graph-theoretic method for the basic reproduction number in continuous time epidemiological models. Journal of Mathematical Biology 59(4):503–516.

- Can we use graph-theoretic methods to improve R_0 calculations?
- Can we use these methods to improve herd-immunity threshold calculations?

Key references

- J. Li, D. Blakeley, R.J. Smith?, The Failure of R_0 (*Comp. Math. Meth. Med*, 2011: 527610)
- I. Yadegari. M. Omidi, S.R. Smith?, The herd-immunity threshold must be updated for multiple vaccine strategies and multiple variants (*Scientific Reports*, 2021, 11:22970)

http://rsmith43.github.io/mysite

